Machine Learning Lecture

Classification and Logistic Regression

Machine Learning

Key Concept

Definition
Classification is a supervised machine learning technique used to predict the **class or category** of a new observation based on its input features.

1 Introduction to Classification

So far, we have studied **Regression**, where the objective is to predict a continuous numerical value.

For example:

- Predict house price = $Rs.50$ lakh
- Predict temperature = $32.5°C$
- Predict salary = $Rs.45{,}000$

However, many real-world problems do not require a numerical value. Instead, we need to determine which **category or class** an observation belongs to.

This is the purpose of **Classification**.

Example

Examples

- Email → Spam / Not Spam
- Student → Pass / Fail
- Transaction → Fraud / Genuine
- Image → Cat / Dog
- Patient → Disease / No Disease

2 Regression vs Classification

The major difference is the type of output predicted by the model.

Feature	Regression	Classification
Output	Continuous value	Class / Category
Example	Salary = $Rs.45{,}000$	Pass / Fail
Target	Numerical	Categorical
Prediction	45.2, 50.7	Class 0, Class 1

Important

Remember

$$\boxed{\text{Regression} \rightarrow \text{Numerical Value}}$$

$$\boxed{\text{Classification} \rightarrow \text{Class / Category}}$$

3 Types of Classification

Classification problems can mainly be divided into:

1. Binary Classification
2. Multiclass Classification

3.1 Binary Classification

Binary classification involves exactly **two possible classes**.

Example

Examples

- Pass / Fail
- Yes / No
- Spam / Not Spam
- Fraud / Genuine
- Disease / No Disease

The classes are often represented using numerical labels:

$$0 \rightarrow \text{Class } 0$$

$$1 \rightarrow \text{Class } 1$$

For example, for student result prediction:

$$0 \rightarrow \text{Fail}$$

$$1 \rightarrow \text{Pass}$$

Key Concept

Binary Classification
Binary classification predicts one of **two possible classes** for a given observation.

3.2 Multiclass Classification

Multiclass classification involves **more than two classes**.

Example

Example: Fruit Classification
Suppose a machine learning model classifies fruits into:

$$\text{Apple}, \quad \text{Banana}, \quad \text{Orange}$$

The model must select one class from these three possible classes.

Other examples include:

- Handwritten digit recognition: 0–9
- Animal classification: Cat, Dog, Horse
- Student grades: A, B, C, D
- Multiple disease classification

Type	**Number of Classes**	**Example**
Binary	2	Pass / Fail
Multiclass	More than 2	Cat / Dog / Horse

4 A Simple Classification Example

Consider a university that wants to predict whether a student will **Pass** or **Fail** based on the number of hours studied.

Suppose the training data is:

Hours Studied	**Result**
1	Fail
2	Fail
3	Fail
4	Pass
5	Pass
6	Pass

We can represent the target using numerical labels:

$$\text{Fail} = 0$$

$$\text{Pass} = 1$$

The model learns the relationship between the input feature **Hours Studied** and the target class.

For a new student, the model should determine whether the student belongs to:

$$\boxed{\text{Fail}}$$

or

$$\boxed{\text{Pass}}$$

5 Why Do We Need Logistic Regression?

A natural question is:

Can we use Linear Regression for classification?

Linear Regression predicts a continuous value:

$$\hat{y} = b_0 + b_1 x$$

Suppose we use:

$$0 = \text{Fail}$$

and

$$1 = \text{Pass}$$

We might expect predictions to lie between 0 and 1.
However, Linear Regression can produce values such as:

$$-0.4, \quad 0.7, \quad 1.3, \quad 2.1$$

Values such as -0.4 or 1.3 cannot directly represent probabilities.

Important

A probability must always lie between:

$$0 \leq P \leq 1$$

Therefore, we need a function that converts the model output into the range 0 to 1.

This leads to **Logistic Regression**.

6 Logistic Regression

Key Concept

Definition
Logistic Regression is a supervised machine learning algorithm used mainly for classification problems. It estimates the probability that an observation belongs to a particular class.

For binary classification, Logistic Regression commonly estimates:

$$P(y = 1)$$

which represents the probability that the observation belongs to **Class 1**.
For example:

$$P(y = 1) = 0.80$$

can be interpreted as an estimated probability of 80% for Class 1.

Important

Although its name contains the word "Regression", Logistic Regression is primarily used for **classification**.

7 Basic Idea of Logistic Regression

Logistic Regression starts with a linear equation similar to Linear Regression.

For one input feature:

$$z = b_0 + b_1 x$$

For multiple input features:

$$z = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_n x_n$$

Here:

- $x_1, x_2, \ldots, x_n$ are input features.
- b_0 is the bias/intercept.
- $b_1, b_2, \ldots, b_n$ are model coefficients.
- z is the weighted sum of the input features.

However, z itself is not the final prediction.
It is passed through the **Sigmoid Function**.

$$z \rightarrow \text{Sigmoid Function} \rightarrow \text{Probability}$$

8 Sigmoid Function

The sigmoid function is defined as:

$$\boxed{\sigma(z) = \frac{1}{1 + e^{-z}}}$$

The main purpose of the sigmoid function is to convert any real-valued input into a value between 0 and 1.

For example:

z	**Sigmoid Output**
-3	0.05
-1	0.27
0	0.50
1	0.73
3	0.95

Therefore:

$$z = -3 \rightarrow 0.05$$

$$z = 0 \rightarrow 0.50$$

$$z = 3 \rightarrow 0.95$$

The output can be interpreted as a probability.

9 Why the Sigmoid Function?

The sigmoid function has an important property:

$$0 < \sigma(z) < 1$$

Therefore, regardless of the value of z, the sigmoid function produces an output between 0 and 1.

This makes it suitable for representing a probability.

Conceptually:

$$\boxed{\text{Linear Combination} \rightarrow \text{Sigmoid} \rightarrow \text{Probability}}$$

10 Probability and Class Prediction

The sigmoid function produces a probability.

For binary classification, we can interpret:

$$P(y = 1)$$

as the probability of belonging to Class 1.

For example:

$$P(y = 1) = 0.80$$

means that the model estimates an 80% probability of Class 1.

Similarly:

$$P(y = 1) = 0.20$$

means that the model estimates a 20% probability of Class 1.

However, we still need to convert this probability into a final class.

This is done using a **Decision Threshold**.

11 Decision Threshold

A commonly used threshold for binary classification is:

$$0.5$$

The decision rule is:

$$P(y = 1) \geq 0.5 \Rightarrow \text{Class } 1$$

$$P(y = 1) < 0.5 \Rightarrow \text{Class } 0$$

For example:

$$P(y = 1) = 0.80$$

Since:

$$0.80 > 0.5$$

the predicted class is:

$$\boxed{\text{Class } 1}$$

Similarly:

$$P(y = 1) = 0.30$$

Since:

$$0.30 < 0.5$$

the predicted class is:

$$\boxed{\text{Class } 0}$$

Important

The threshold is not the same thing as the probability.

- **Probability** tells us how strongly the model favors a class.
- **Threshold** is the rule used to convert that probability into a final class.

12 Complete Logistic Regression Flow

The complete process can be represented as:

$$\boxed{X \rightarrow z \rightarrow \text{Sigmoid} \rightarrow P(y = 1) \rightarrow \text{Threshold} \rightarrow \text{Class}}$$

where:

$$z = b_0 + b_1 x_1 + \cdots + b_n x_n$$

and:

$$P(y = 1) = \frac{1}{1 + e^{-z}}$$

Finally:

$$P(y = 1) \geq 0.5 \Rightarrow \text{Class } 1$$

$$P(y = 1) < 0.5 \Rightarrow \text{Class } 0$$

13 Complete Example: Student Pass/Fail Prediction

Consider a model that predicts whether a student will pass based on hours studied.

Suppose the trained model produces:

$$z = -5 + 1.2x$$

where x represents the number of hours studied.

For a student who studies for 5 hours:

$$z = -5 + (1.2)(5)$$

$$z = 1$$

Now apply the sigmoid function:

$$P(y = 1) = \frac{1}{1 + e^{-1}}$$

$$P(y = 1) \approx 0.73$$

Therefore:

$$P(\text{Pass}) \approx 0.73$$

or approximately:

$$\boxed{73\%}$$

Using a threshold of 0.5:

$$0.73 > 0.5$$

Therefore:

$$\boxed{\text{Predicted Class} = \text{Pass}}$$

14 Logistic Regression with Multiple Features

Logistic Regression can use more than one input feature.

For example, student performance may depend on:

- Hours studied
- Attendance
- Assignment score

The model first calculates:

$$z = b_0 + b_1x_1 + b_2x_2 + b_3x_3$$

The value of z is then passed through the sigmoid function:

$$P(y = 1) = \frac{1}{1 + e^{-z}}$$

Suppose the model produces:

$$P(\text{Pass}) = 0.87$$

Since:

$$0.87 > 0.5$$

the final prediction is:

$$\boxed{\text{Pass}}$$

15 Key Terms

Term	**Meaning**
Classification	Predicting a class or category
Binary Classification	Classification with two classes
Multiclass Classification	Classification with more than two classes
Logistic Regression	Classification algorithm based on probability
Coefficient	Weight learned for an input feature
Bias	Intercept of the model
z	Weighted sum of input features
Sigmoid Function	Converts z into a value between 0 and 1
Probability	Estimated likelihood of a class
Threshold	Rule used to convert probability into a class

16 Summary

Key Concept

- Classification is used to predict a **class or category**.
- Binary classification has **two classes**.
- Multiclass classification has **more than two classes**.
- Logistic Regression is mainly used for **classification**.
- Logistic Regression first calculates a weighted sum:

$$z = b_0 + b_1 x_1 + \cdots + b_n x_n$$

- The sigmoid function converts z into a value between 0 and 1:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- The sigmoid output is interpreted as a probability.
- A decision threshold is then used to obtain the final class.

Features → Weighted Sum → Sigmoid → Probability → Threshold → Class